

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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Department:	CSE (CS)	Date:	27-08-2026

ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I Shaik Basheer (192465049) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Basheer

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Comparison of L1, L2 and L3 Cache

The three cache levels are not simply three copies of the same memory. They form a latency–capacity–cost trade-off. L1 is closest to the execution units and is normally the first cache checked; L2 is larger and slower; L3 is typically larger again and may be shared across cores. AMD documentation describes internal caches as the closest hierarchy levels and notes that higher-capacity levels are slower than lower levels. [\[cite turn0search36 \]](#)

Comparative Analysis

Parameter	L1 Cache	L2 Cache	L3 Cache	Architectural meaning
Relative latency	Lowest	Low	Higher than L1/L2	Misses become progressively more expensive
Capacity	Smallest	Larger	Largest of the three	Larger working sets can be retained
Bandwidth potential	Highest local access rate	High	High, but shared/interconnect dependent	Lower levels can feed execution units quickly
Typical sharing	Usually private to a core	Usually private or core-local	Often shared by multiple cores	L3 can improve inter-core data reuse
Main purpose	Immediate instructions/data	Extend fast working set	Reduce DRAM traffic	Balance speed and capacity

Why the Difference Matters

- A high L1 hit rate keeps critical loads close to the execution pipeline and avoids deeper-cache penalties.
- L2 absorbs many L1 misses without immediately going to the shared last-level cache or DRAM.
- L3 is valuable for large working sets and multicore sharing, but its larger size and sharing can increase access and contention costs.
- Cache misses can increase processor stall cycles; AMD/Xilinx performance material explicitly links cache misses and memory stalls to reduced effective bandwidth. [\[cite turn0search39 \]](#)

Critical Judgment

There is no universally 'best' cache level. For latency-sensitive code, L1 is most valuable; for a moderate working set, L2 can provide a strong compromise; for multicore workloads with larger shared datasets, L3 can reduce expensive DRAM traffic. Therefore, increasing L3 alone is not a guaranteed performance solution: locality, hit rate, sharing and workload size determine the benefit.

Practical Evidence

A current AMD EPYC example lists 32 KB instruction and 32 KB data L1 per core, 512 KB–1 MB L2 per core depending on model, and tens to hundreds of MB of L3 per processor. The same guide lists memory speeds up to 5,200 MT/s for one tested processor, illustrating that cache capacity and main-memory bandwidth are separate design dimensions. [\[cite turn0search38 \]](#)

Conclusion

The optimal hierarchy uses L1 for the hottest data, L2 for the next working set and L3 as a larger/shared buffer before DRAM. Performance depends more on effective hit behavior and locality than on cache size alone.

2. SRAM versus DRAM

SRAM and DRAM both store volatile data, but their circuit organization creates a fundamental trade-off: SRAM is used where very low access latency is worth the larger area and lower density, while DRAM is used where large capacity and economical main memory are required.

Comparison

Parameter	SRAM	DRAM	Design consequence
Storage principle	Bistable cell	Capacitor-based dynamic cell	DRAM requires refresh
Latency	Very low	Higher than SRAM	SRAM suits cache; DRAM suits main memory
Density	Lower	Higher	DRAM provides much more capacity per area
Refresh	Not required like DRAM	Required	Refresh consumes time/energy
Typical role	CPU cache	Main memory	Creates hierarchy rather than direct substitution
Cost/bit	Higher	Lower	Large systems favor DRAM for capacity

Latency, Bandwidth and Throughput

Latency is the delay for an individual access; bandwidth is the amount of data transferable per unit time; throughput is the useful work the system sustains. A memory can have high transfer bandwidth and still have higher access latency than SRAM. Modern DRAM demonstrates why bandwidth matters: Micron states that DDR5 provides twice the effective bandwidth of DDR4 in its product portfolio. ^{[7] cite[7] turn0search13[7]}

Critical Judgment

Replacing all DRAM with SRAM would reduce latency but would be impractical for large-capacity systems because SRAM is area- and cost-intensive. Replacing cache SRAM with DRAM would save area but would increase the latency seen by the processor. The better engineering solution is hierarchical: use SRAM selectively for high-value, frequently accessed data and DRAM for the much larger working set.

Conclusion

SRAM is the better technology for cache-level latency; DRAM is the better technology for economical high-capacity main memory. The technologies are complementary, not competitors for the same hierarchy position.

3. Virtual Memory with Paging versus Cache Memory

Cache memory and virtual-memory paging address different problems. Cache reduces the average latency of processor accesses by keeping copies of frequently used data close to the CPU. Paging manages a virtual address space by moving fixed-size pages between physical memory and backing storage when necessary.

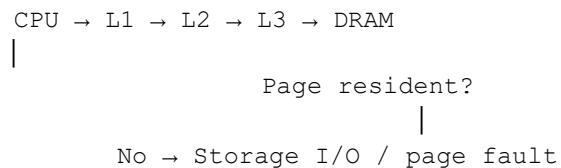
Direct Comparison

Aspect	Cache	Paging / Virtual Memory	Performance implication
Primary objective	Reduce average memory access latency	Manage address space and physical-memory capacity	Different goals
Normal fast path	Cache hit	Page resident in DRAM	Both can be fast when locality is good
Failure event	Cache miss	Page fault	Page fault is much more expensive
Data movement	Cache-line granularity	Page granularity	Paging can move much larger blocks
Best optimization	Locality, cache-aware access, prefetching	Adequate DRAM, working-set control, fewer faults	Different tuning methods
Throughput effect	Fewer CPU stalls	Fewer storage I/O events	Both can improve system throughput when tuned

Critical Insight

It is misleading to say that paging is a 'slower cache.' A cache is a performance accelerator inside the memory hierarchy, whereas virtual memory is an address-management and protection mechanism whose backing storage can extend the apparent memory capacity. A page fault may require storage I/O, making its penalty dramatically larger than a cache miss.

Architecture



Design Judgment

For a database or scientific workload, adding cache capacity cannot compensate for severe DRAM shortage if the system is constantly paging. Conversely, adding DRAM does not remove the need for cache because processor-to-DRAM latency remains much larger than cache access. The correct solution is to optimize both locality and working-set residency.

Conclusion

Cache primarily improves speed; paging primarily manages capacity and isolation. Maximum throughput comes from keeping the active working set in DRAM and maximizing cache locality so that both cache misses and page faults remain low.

4. NAND Flash versus DRAM

NAND Flash and DRAM occupy fundamentally different hierarchy positions. DRAM is volatile working memory designed for frequent reads and writes, while NAND Flash is non-volatile, high-density storage. Treating them as interchangeable would ignore the latency, endurance and persistence trade-offs.

Comparison

Parameter	DRAM	NAND Flash	Architectural decision
Volatility	Volatile	Non-volatile	Flash preserves data without power
Access behavior	Designed for frequent random working-memory access	Storage-oriented; controller/firmware involved	DRAM is closer to CPU
Capacity density	High memory capacity, but lower density than NAND storage	Very high density	NAND is preferred for persistent mass storage
Latency	Much lower for workingmemory access	Higher and workload/controller dependent	Do not use raw NAND as a direct DRAM replacement
Write characteristics	Frequent writes supported in normal operation	Program/erase limits and wear management	Flash needs ECC, wear leveling and bad-block management
Typical role	Main memory	SSD/embedded storage	Different hierarchy layers

Evidence

Micron notes that NAND requires controller functions such as ECC, bad-block management and wear leveling. It also distinguishes high-density NAND for data-heavy applications and notes that higher-density cell types can trade performance and endurance for density. [1]

As an example of current NAND capability, Micron reports G8 NAND I/O speed of up to 2.4 GB/s and capacities up to 1 Tb per die and 2 TB per package. These figures describe storage-device capability, not DRAM-equivalent randomaccess latency. [8]

Critical Judgment

NAND is the better choice when persistence, density and cost per stored bit dominate. DRAM is the better choice when the processor needs a repeatedly accessed, writable working set with low access latency. Therefore, the strongest hierarchy uses both: DRAM for active computation and NAND for persistent programs, models and datasets.

Conclusion

DRAM should remain the main working memory, while NAND Flash should provide high-density persistent storage. The performance gap is precisely why a hierarchy is necessary.

5. MRAM versus RRAM

MRAM and RRAM are non-volatile memory technologies with different physical mechanisms and engineering tradeoffs. The comparison should therefore avoid declaring one technology universally superior; the correct choice depends on access speed, endurance, density, energy and the target workload.

Comparison

Aspect	MRAM	RRAM / ReRAM	Critical interpretation
State mechanism	Magnetic state	Resistance state	Different device physics create different scaling and reliability trade-offs
Non-volatility	Yes	Yes	Both retain information without continuous power
Access performance	Potentially fast; studied for cache and memory applications	Potentially fast and highly dense; performance depends on cell/array design	Actual product performance must be measured at system level
Endurance	Generally strong for many MRAM approaches	Varies by device and switching mechanism	Workload write intensity matters
Density/scaling	Good potential, but magnetic structures have area constraints	Strong density potential	RRAM is attractive for dense arrays
Energy	Can reduce standby energy because data is retained	Can be energy-efficient in suitable architectures	Write energy and peripheral circuitry must be included
Applications	Embedded memory, cache research, persistent memory	Embedded memory and compute-in-memory research	Technology maturity and workload determine suitability

Evidence and Research Context

IEEE literature includes direct comparisons of STT- and SOT-MRAM cells for cache applications, showing that MRAM performance must be evaluated at the cell/cache architecture level rather than by a single headline number. IEEE research also reports heterogeneous RRAM/SRAM/MRAM benchmarking for neuralnetwork applications, demonstrating that hybrid memory organizations can be more meaningful than a one-technology replacement.

Critical Judgment

If the priority is a fast, persistent memory suitable for embedded/cache-oriented designs, MRAM is a strong candidate. If density and emerging compute-in-memory or large-array potential are more important, RRAM can be attractive. However, RRAM performance is highly dependent on device variability, array architecture and peripheral circuitry, so claiming that RRAM is always faster or more energy-efficient would be technically unjustified.

Recommended Selection

Priority	Preferred direction	Reason
Fast persistent embedded/cache memory	MRAM	Strong fit for low-latency nonvolatile memory research and products
High-density emerging memory arrays	RRAM	Strong density and resistive-array potential
AI/compute-in-memory research	Hybrid RRAM/MRAM/SRAM	Allows each technology to be used where its characteristics are strongest
Lowest system risk	Mature, qualified technology selected from measured datasheets	Device-level claims alone are insufficient

Conclusion

MRAM and RRAM should be viewed as complementary emerging memory choices. MRAM is attractive for fast persistent memory, while RRAM is attractive for dense arrays and emerging compute-in-memory designs. The final decision must use measured access time, write energy, endurance and array-level throughput for the intended application.

Cross-Question Engineering Findings

Question	Main trade-off	Best engineering conclusion
L1/L2/L3	Latency vs capacity vs sharing	Use all levels; optimize locality and hit rate
SRAM/DRAM	Latency/cost vs capacity	SRAM for cache, DRAM for main memory
Cache/Paging	Speed optimization vs capacity/address management	Optimize both cache locality and DRAM working-set residency
NAND/DRAM	Persistence/density vs low-latency working memory	DRAM for active data, NAND for persistent storage
MRAM/RRAM	Non-volatility vs density/endurance/energy	Choose by workload; hybrid designs can be stronger than replacement